

adhd 8812 1

ADHD

```
packages <- c(
  "readxl", "dplyr", "tidyr", "stringr",
  "lavaan", "semTools", "psych", "broom"
)

to_install <- packages[!packages %in% rownames(installed.packages())]
if(length(to_install) > 0) install.packages(to_install)

library(readxl)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(stringr)
library(lavaan)
```

This is lavaan 0.6-21

lavaan is FREE software! Please report any bugs.

```
library(semTools)
```

#####

This is semTools 0.5-8

All users of R (or SEM) are invited to submit functions or ideas for functions.

#####

```
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:semTools':

reliability, skew

The following object is masked from 'package:lavaan':

cor2cov

```
library(broom)
library(dplyr)

data_path <- "/Users/yangsiyutong/Desktop/QuestionPro-SR-RawData-4077617-04-12-2026-T0
knitr::opts_chunk$set(
  message = FALSE,
  warning = FALSE
)

raw <- read_excel(
  data_path,
  sheet = "原始数据",
  .name_repair = "minimal"
)

dat <- tibble(
  id = raw[[1]],
  sex = raw[[10]],
  age = raw[[11]],

  multi_partner = raw[[12]],

  mate1 = raw[[25]],
  mate2 = raw[[26]],
  mate3 = raw[[27]],
  mate4 = raw[[28]],
  mate5 = raw[[29]],

  adhd1 = raw[[33]],
  adhd2 = raw[[34]],
  adhd3 = raw[[35]],
  adhd4 = raw[[36]],
  adhd5 = raw[[37]],
  adhd6 = raw[[38]],
  adhd7 = raw[[39]],
  adhd8 = raw[[40]],
  adhd9 = raw[[41]],
  adhd10 = raw[[42]],
  adhd11 = raw[[43]],
  adhd12 = raw[[44]],
  adhd13 = raw[[45]],
  adhd14 = raw[[46]],
  adhd15 = raw[[47]],
  adhd16 = raw[[48]],
  adhd17 = raw[[49]],
  adhd18 = raw[[50]]
```

```
)

adhd_map <- c(
  "从不" = 1,
  "很少" = 2,
  "有时" = 3,
  "经常" = 4,
  "非常经常" = 5
)

recode_adhd <- function(x){
  x <- str_squish(as.character(x))
  x_num <- suppressWarnings(as.numeric(x))
  x_lab <- as.numeric(unname(adhd_map[x]))
  ifelse(!is.na(x_num), x_num, x_lab)
}

recode_inter <- function(x){
  x <- str_squish(as.character(x))
  case_when(
    str_detect(x, "^B") ~ 1,
    str_detect(x, "^A") ~ 0,
    TRUE ~ NA_real_
  )
}

row_mean_min <- function(data, vars, min_items = 1){
  x <- data[, vars] %>%
    mutate(across(everything(), as.numeric)) %>%
    as.matrix()

  valid_n <- rowSums(!is.na(x))
  out <- rowMeans(x, na.rm = TRUE)
  out[valid_n < min_items] <- NA_real_
  out
}

adhd_items <- paste0("adhd", 1:18)

inatt_items <- c(
  "adhd1", "adhd2", "adhd3", "adhd4",
  "adhd7", "adhd8", "adhd9", "adhd10", "adhd11"
)

hyper_items <- c(
  "adhd5", "adhd6",
  "adhd12", "adhd13", "adhd14", "adhd15",
  "adhd16", "adhd17", "adhd18"
)

inter_items <- paste0("inter", 1:5)
intra_items <- paste0("intra", 1:5)

dat <- dat %>%
```

```

mutate(
  across(all_of(adhd_items), recode_adhd),

  inter1 = recode_inter(mate1),
  inter2 = recode_inter(mate2),
  inter3 = recode_inter(mate3),
  inter4 = recode_inter(mate4),
  inter5 = recode_inter(mate5),

  intra1 = 1 - inter1,
  intra2 = 1 - inter2,
  intra3 = 1 - inter3,
  intra4 = 1 - inter4,
  intra5 = 1 - inter5
)

dat <- dat %>%
  mutate(
    adhd_total = row_mean_min(., adhd_items, min_items = 14),
    adhd_inattention = row_mean_min(., inatt_items, min_items = 7),
    adhd_hyperimpulsive = row_mean_min(., hyper_items, min_items = 7),

    inter_score = row_mean_min(., inter_items, min_items = 3),
    intra_score = row_mean_min(., intra_items, min_items = 3),

    inter_majority = if_else(inter_score >= 0.60, 1, 0, missing = NA_real_),
    intra_majority = if_else(intra_score >= 0.60, 1, 0, missing = NA_real_)
  )

str(dat[, adhd_items])

```

```

tibble [297 × 18] (S3: tbl_df/tbl/data.frame)
 $ adhd1 : num [1:297] 2 4 4 2 5 4 4 3 3 5 ...
 $ adhd2 : num [1:297] 2 5 4 3 2 3 3 4 3 4 ...
 $ adhd3 : num [1:297] 2 3 3 2 5 2 5 3 3 3 ...
 $ adhd4 : num [1:297] 2 5 4 4 4 4 2 4 4 4 ...
 $ adhd5 : num [1:297] 3 4 3 2 3 4 3 5 3 4 ...
 $ adhd6 : num [1:297] 2 5 3 2 4 5 3 5 5 5 ...
 $ adhd7 : num [1:297] 4 5 4 3 4 2 4 5 3 2 ...
 $ adhd8 : num [1:297] 3 4 4 3 4 4 5 3 2 3 ...
 $ adhd9 : num [1:297] 4 3 4 2 5 2 4 4 5 4 ...
 $ adhd10: num [1:297] 2 3 5 2 4 2 4 5 4 4 ...
 $ adhd11: num [1:297] 4 5 4 2 4 3 5 3 4 4 ...
 $ adhd12: num [1:297] 2 4 4 1 4 3 3 4 4 4 ...
 $ adhd13: num [1:297] 3 5 2 1 5 4 4 4 5 5 ...
 $ adhd14: num [1:297] 2 4 3 1 3 3 4 3 4 3 ...
 $ adhd15: num [1:297] 3 4 3 2 4 4 3 5 2 4 ...
 $ adhd16: num [1:297] 3 5 5 2 4 4 5 4 3 4 ...
 $ adhd17: num [1:297] 2 4 3 2 4 4 4 4 4 4 ...
 $ adhd18: num [1:297] 2 5 4 2 4 4 5 4 4 3 ...

```

```
summary(dat$adhd_total)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
1.000	2.278	3.167	3.042	3.889	5.000	1

```
summary(dat$inter_score)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.0000	0.4000	0.8000	0.6581	1.0000	1.0000	1

```
summary(dat$intra_score)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.0000	0.0000	0.2000	0.3419	0.6000	1.0000	1

```
table(dat$inter_majority, useNA = "ifany")
```

0	1	<NA>
91	205	1

```
table(dat$intra_majority, useNA = "ifany")
```

0	1	<NA>
205	91	1

##CFA model: ADHD as a latent variable

```
dat_cfa <- dat %>%
  select(all_of(adhd_items)) %>%
  drop_na() %>%
  mutate(across(everything(), ordered))

adhd_cfa_total <- paste0(
  "ADHD =~ ",
  paste(adhd_items, collapse = " + ")
)

fit_cfa_total <- cfa(
  adhd_cfa_total,
  data = dat_cfa,
  ordered = adhd_items,
  estimator = "WLSMV",
  parameterization = "theta",
  std.lv = TRUE
)

summary(
  fit_cfa_total,
  fit.measures = TRUE,
  standardized = TRUE
)
```

lavaan 0.6-21 ended normally after 31 iterations

Estimator	DWLS
Optimization method	NLMINB
Number of model parameters	90
Number of observations	291

Model Test User Model:

	Standard	Scaled
Test Statistic	271.053	422.677
Degrees of freedom	135	135
P-value (Unknown)	NA	0.000
Scaling correction factor		0.733
Shift parameter		53.081
simple second-order correction		

Model Test Baseline Model:

Test statistic	53670.987	15373.277
Degrees of freedom	153	153
P-value	NA	0.000
Scaling correction factor		3.516

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.997	0.981
Tucker-Lewis Index (TLI)	0.997	0.979
Robust Comparative Fit Index (CFI)		0.921
Robust Tucker-Lewis Index (TLI)		0.910

Root Mean Square Error of Approximation:

RMSEA	0.059	0.086
90 Percent confidence interval - lower	0.049	0.077
90 Percent confidence interval - upper	0.069	0.095
P-value H ₀ : RMSEA ≤ 0.050	0.074	0.000
P-value H ₀ : RMSEA ≥ 0.080	0.000	0.851
Robust RMSEA		0.098
90 Percent confidence interval - lower		0.088
90 Percent confidence interval - upper		0.109
P-value H ₀ : Robust RMSEA ≤ 0.050		0.000
P-value H ₀ : Robust RMSEA ≥ 0.080		0.998

Standardized Root Mean Square Residual:

SRMR	0.045	0.045
------	-------	-------

Parameter Estimates:

Parameterization	Theta
Standard errors	Robust.sem

Information

Expected

Information saturated (h1) model

Unstructured

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
ADHD =~						
adhd1	1.203	0.094	12.852	0.000	1.203	0.769
adhd2	1.096	0.090	12.128	0.000	1.096	0.739
adhd3	1.467	0.109	13.493	0.000	1.467	0.826
adhd4	1.304	0.096	13.622	0.000	1.304	0.794
adhd5	1.399	0.102	13.672	0.000	1.399	0.813
adhd6	1.168	0.089	13.105	0.000	1.168	0.760
adhd7	1.490	0.110	13.605	0.000	1.490	0.830
adhd8	1.343	0.098	13.726	0.000	1.343	0.802
adhd9	1.696	0.122	13.893	0.000	1.696	0.861
adhd10	1.358	0.097	14.011	0.000	1.358	0.805
adhd11	1.407	0.100	14.059	0.000	1.407	0.815
adhd12	1.532	0.113	13.569	0.000	1.532	0.837
adhd13	1.553	0.116	13.342	0.000	1.553	0.841
adhd14	1.490	0.108	13.830	0.000	1.490	0.830
adhd15	1.149	0.091	12.633	0.000	1.149	0.754
adhd16	1.337	0.103	13.002	0.000	1.337	0.801
adhd17	1.305	0.098	13.296	0.000	1.305	0.794
adhd18	0.968	0.083	11.712	0.000	0.968	0.695

Thresholds:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
adhd1 t1	-1.809	0.146	-12.364	0.000	-1.809	-1.157
adhd1 t2	-0.872	0.120	-7.278	0.000	-0.872	-0.557
adhd1 t3	0.209	0.116	1.805	0.071	0.209	0.134
adhd1 t4	2.323	0.154	15.092	0.000	2.323	1.485
adhd2 t1	-1.793	0.142	-12.658	0.000	-1.793	-1.209
adhd2 t2	-0.542	0.110	-4.939	0.000	-0.542	-0.365
adhd2 t3	0.342	0.111	3.089	0.002	0.342	0.230
adhd2 t4	1.620	0.135	11.966	0.000	1.620	1.092
adhd3 t1	-1.831	0.157	-11.693	0.000	-1.831	-1.031
adhd3 t2	-0.409	0.129	-3.168	0.002	-0.409	-0.230
adhd3 t3	0.715	0.139	5.152	0.000	0.715	0.403
adhd3 t4	2.314	0.172	13.460	0.000	2.314	1.303
adhd4 t1	-1.901	0.149	-12.715	0.000	-1.901	-1.157
adhd4 t2	-0.677	0.120	-5.633	0.000	-0.677	-0.412
adhd4 t3	0.292	0.122	2.382	0.017	0.292	0.178
adhd4 t4	1.957	0.151	12.987	0.000	1.957	1.191
adhd5 t1	-2.048	0.152	-13.433	0.000	-2.048	-1.191
adhd5 t2	-0.856	0.130	-6.604	0.000	-0.856	-0.498
adhd5 t3	0.472	0.128	3.688	0.000	0.472	0.275
adhd5 t4	1.773	0.145	12.267	0.000	1.773	1.031
adhd6 t1	-2.068	0.155	-13.332	0.000	-2.068	-1.345
adhd6 t2	-0.677	0.117	-5.793	0.000	-0.677	-0.440
adhd6 t3	0.273	0.114	2.401	0.016	0.273	0.178
adhd6 t4	1.779	0.133	13.412	0.000	1.779	1.157
adhd7 t1	-2.375	0.177	-13.388	0.000	-2.375	-1.324
adhd7 t2	-0.824	0.134	-6.140	0.000	-0.824	-0.459
adhd7 t3	0.429	0.134	3.197	0.001	0.429	0.239

adhd7 t4	2.075	0.171	12.120	0.000	2.075	1.157
adhd8 t1	-1.965	0.153	-12.858	0.000	-1.965	-1.174
adhd8 t2	-0.785	0.125	-6.289	0.000	-0.785	-0.469
adhd8 t3	0.239	0.125	1.916	0.055	0.239	0.143
adhd8 t4	2.487	0.175	14.216	0.000	2.487	1.485
adhd9 t1	-2.119	0.174	-12.204	0.000	-2.119	-1.076
adhd9 t2	-0.756	0.144	-5.241	0.000	-0.756	-0.384
adhd9 t3	0.630	0.149	4.223	0.000	0.630	0.320
adhd9 t4	2.489	0.167	14.898	0.000	2.489	1.264
adhd10 t1	-2.165	0.164	-13.188	0.000	-2.165	-1.284
adhd10 t2	-0.663	0.125	-5.314	0.000	-0.663	-0.393
adhd10 t3	0.479	0.127	3.756	0.000	0.479	0.284
adhd10 t4	2.165	0.160	13.500	0.000	2.165	1.284
adhd11 t1	-2.216	0.164	-13.530	0.000	-2.216	-1.284
adhd11 t2	-0.859	0.128	-6.710	0.000	-0.859	-0.498
adhd11 t3	0.413	0.130	3.187	0.001	0.413	0.239
adhd11 t4	2.397	0.180	13.339	0.000	2.397	1.389
adhd12 t1	-1.758	0.147	-11.960	0.000	-1.758	-0.961
adhd12 t2	-0.602	0.135	-4.442	0.000	-0.602	-0.329
adhd12 t3	0.470	0.137	3.429	0.001	0.470	0.257
adhd12 t4	2.349	0.175	13.421	0.000	2.349	1.284
adhd13 t1	-2.076	0.168	-12.327	0.000	-2.076	-1.124
adhd13 t2	-0.608	0.135	-4.508	0.000	-0.608	-0.329
adhd13 t3	0.377	0.138	2.722	0.006	0.377	0.204
adhd13 t4	2.300	0.161	14.276	0.000	2.300	1.245
adhd14 t1	-2.016	0.161	-12.495	0.000	-2.016	-1.124
adhd14 t2	-0.606	0.132	-4.587	0.000	-0.606	-0.338
adhd14 t3	0.461	0.135	3.416	0.001	0.461	0.257
adhd14 t4	2.492	0.174	14.295	0.000	2.492	1.389
adhd15 t1	-2.048	0.153	-13.398	0.000	-2.048	-1.345
adhd15 t2	-0.613	0.115	-5.329	0.000	-0.613	-0.403
adhd15 t3	0.405	0.113	3.582	0.000	0.405	0.266
adhd15 t4	1.616	0.137	11.790	0.000	1.616	1.061
adhd16 t1	-2.017	0.152	-13.297	0.000	-2.017	-1.209
adhd16 t2	-0.384	0.124	-3.109	0.002	-0.384	-0.230
adhd16 t3	0.504	0.126	4.011	0.000	0.504	0.302
adhd16 t4	2.210	0.164	13.441	0.000	2.210	1.324
adhd17 t1	-1.848	0.149	-12.376	0.000	-1.848	-1.124
adhd17 t2	-0.078	0.121	-0.645	0.519	-0.078	-0.047
adhd17 t3	0.586	0.124	4.708	0.000	0.586	0.356
adhd17 t4	2.360	0.164	14.351	0.000	2.360	1.435
adhd18 t1	-1.733	0.136	-12.752	0.000	-1.733	-1.245
adhd18 t2	-0.407	0.104	-3.931	0.000	-0.407	-0.293
adhd18 t3	0.198	0.103	1.931	0.054	0.198	0.143
adhd18 t4	1.814	0.132	13.772	0.000	1.814	1.303

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.adhd1	1.000				1.000	0.409
.adhd2	1.000				1.000	0.454
.adhd3	1.000				1.000	0.317
.adhd4	1.000				1.000	0.370
.adhd5	1.000				1.000	0.338
.adhd6	1.000				1.000	0.423

.adhd7	1.000	1.000	0.311
.adhd8	1.000	1.000	0.357
.adhd9	1.000	1.000	0.258
.adhd10	1.000	1.000	0.352
.adhd11	1.000	1.000	0.336
.adhd12	1.000	1.000	0.299
.adhd13	1.000	1.000	0.293
.adhd14	1.000	1.000	0.311
.adhd15	1.000	1.000	0.431
.adhd16	1.000	1.000	0.359
.adhd17	1.000	1.000	0.370
.adhd18	1.000	1.000	0.516
ADHD	1.000	1.000	1.000

Scales y*:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
adhd1	0.639				0.639	1.000
adhd2	0.674				0.674	1.000
adhd3	0.563				0.563	1.000
adhd4	0.609				0.609	1.000
adhd5	0.582				0.582	1.000
adhd6	0.650				0.650	1.000
adhd7	0.557				0.557	1.000
adhd8	0.597				0.597	1.000
adhd9	0.508				0.508	1.000
adhd10	0.593				0.593	1.000
adhd11	0.579				0.579	1.000
adhd12	0.546				0.546	1.000
adhd13	0.541				0.541	1.000
adhd14	0.557				0.557	1.000
adhd15	0.657				0.657	1.000
adhd16	0.599				0.599	1.000
adhd17	0.608				0.608	1.000
adhd18	0.719				0.719	1.000

```
fitMeasures(
  fit_cfa_total,
  c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "srmr")
)
```

chisq	df	pvalue	cfi	tli	rmsea	srmr
271.053	135.000	NA	0.997	0.997	0.059	0.045

```
semTools::reliability(fit_cfa_total)
```

	ADHD
alpha	0.9635576
alpha.ord	0.9682948
omega	0.9622748
omega2	0.9622748
omega3	0.9731491
avevar	0.6489281

Structural model: latent ADHD predicting intersexual-display binary outcome

```

dat_sem_inter <- dat %>%
  select(all_of(adhd_items), inter_majority) %>%
  drop_na() %>%
  mutate(across(everything(), ordered))

sem_inter_model <- paste0(
  "ADHD =~ ",
  paste(adhd_items, collapse = " + "),
  "\ninter_majority ~ ADHD"
)

fit_sem_inter <- sem(
  sem_inter_model,
  data = dat_sem_inter,
  ordered = c(adhd_items, "inter_majority"),
  estimator = "WLSMV",
  parameterization = "theta",
  std.lv = TRUE
)

summary(
  fit_sem_inter,
  fit.measures = TRUE,
  standardized = TRUE
)

```

lavaan 0.6-21 ended normally after 32 iterations

Estimator	DWLS
Optimization method	NLMINB
Number of model parameters	92
Number of observations	291

Model Test User Model:

	Standard	Scaled
Test Statistic	309.880	480.633
Degrees of freedom	152	152
P-value (Unknown)	NA	0.000
Scaling correction factor		0.738
Shift parameter		60.747
simple second-order correction		

Model Test Baseline Model:

Test statistic	53857.185	15701.999
Degrees of freedom	171	171
P-value	NA	0.000
Scaling correction factor		3.457

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.997	0.979
Tucker–Lewis Index (TLI)	0.997	0.976
Robust Comparative Fit Index (CFI)		0.902
Robust Tucker–Lewis Index (TLI)		0.890

Root Mean Square Error of Approximation:

RMSEA	0.060	0.086
90 Percent confidence interval – lower	0.050	0.078
90 Percent confidence interval – upper	0.069	0.095
P-value H ₀ : RMSEA ≤ 0.050	0.046	0.000
P-value H ₀ : RMSEA ≥ 0.080	0.000	0.888
Robust RMSEA		0.104
90 Percent confidence interval – lower		0.094
90 Percent confidence interval – upper		0.114
P-value H ₀ : Robust RMSEA ≤ 0.050		0.000
P-value H ₀ : Robust RMSEA ≥ 0.080		1.000

Standardized Root Mean Square Residual:

SRMR	0.053	0.053
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Parameter Estimates:

Parameterization	Theta
Standard errors	Robust.sem
Information	Expected
Information saturated (h1) model	Unstructured

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
ADHD =~						
adhd1	1.204	0.094	12.863	0.000	1.204	0.769
adhd2	1.095	0.090	12.133	0.000	1.095	0.739
adhd3	1.463	0.108	13.499	0.000	1.463	0.826
adhd4	1.304	0.096	13.642	0.000	1.304	0.794
adhd5	1.399	0.102	13.673	0.000	1.399	0.814
adhd6	1.168	0.089	13.102	0.000	1.168	0.760
adhd7	1.489	0.109	13.605	0.000	1.489	0.830
adhd8	1.344	0.098	13.730	0.000	1.344	0.802
adhd9	1.696	0.122	13.868	0.000	1.696	0.861
adhd10	1.356	0.097	14.019	0.000	1.356	0.805
adhd11	1.405	0.100	14.052	0.000	1.405	0.815
adhd12	1.529	0.113	13.570	0.000	1.529	0.837
adhd13	1.550	0.116	13.357	0.000	1.550	0.840
adhd14	1.488	0.108	13.812	0.000	1.488	0.830
adhd15	1.150	0.091	12.631	0.000	1.150	0.755
adhd16	1.341	0.103	12.998	0.000	1.341	0.802
adhd17	1.305	0.098	13.281	0.000	1.305	0.794
adhd18	0.975	0.083	11.748	0.000	0.975	0.698

Regressions:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
inter_majority ~						
ADHD	0.282	0.082	3.441	0.001	0.282	0.271

Thresholds:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
adhd1 t1	-1.810	0.146	-12.363	0.000	-1.810	-1.157
adhd1 t2	-0.872	0.120	-7.278	0.000	-0.872	-0.557
adhd1 t3	0.210	0.116	1.805	0.071	0.210	0.134
adhd1 t4	2.324	0.154	15.130	0.000	2.324	1.485
adhd2 t1	-1.793	0.142	-12.668	0.000	-1.793	-1.209
adhd2 t2	-0.542	0.110	-4.938	0.000	-0.542	-0.365
adhd2 t3	0.342	0.111	3.089	0.002	0.342	0.230
adhd2 t4	1.619	0.135	11.972	0.000	1.619	1.092
adhd3 t1	-1.828	0.156	-11.703	0.000	-1.828	-1.031
adhd3 t2	-0.408	0.129	-3.168	0.002	-0.408	-0.230
adhd3 t3	0.713	0.138	5.153	0.000	0.713	0.403
adhd3 t4	2.310	0.171	13.473	0.000	2.310	1.303
adhd4 t1	-1.901	0.149	-12.724	0.000	-1.901	-1.157
adhd4 t2	-0.677	0.120	-5.633	0.000	-0.677	-0.412
adhd4 t3	0.292	0.122	2.382	0.017	0.292	0.178
adhd4 t4	1.957	0.151	12.997	0.000	1.957	1.191
adhd5 t1	-2.048	0.152	-13.454	0.000	-2.048	-1.191
adhd5 t2	-0.856	0.130	-6.604	0.000	-0.856	-0.498
adhd5 t3	0.472	0.128	3.688	0.000	0.472	0.275
adhd5 t4	1.773	0.144	12.279	0.000	1.773	1.031
adhd6 t1	-2.068	0.155	-13.325	0.000	-2.068	-1.345
adhd6 t2	-0.677	0.117	-5.791	0.000	-0.677	-0.440
adhd6 t3	0.273	0.114	2.401	0.016	0.273	0.178
adhd6 t4	1.779	0.133	13.416	0.000	1.779	1.157
adhd7 t1	-2.375	0.177	-13.388	0.000	-2.375	-1.324
adhd7 t2	-0.824	0.134	-6.140	0.000	-0.824	-0.459
adhd7 t3	0.429	0.134	3.197	0.001	0.429	0.239
adhd7 t4	2.075	0.171	12.121	0.000	2.075	1.157
adhd8 t1	-1.966	0.153	-12.867	0.000	-1.966	-1.174
adhd8 t2	-0.785	0.125	-6.289	0.000	-0.785	-0.469
adhd8 t3	0.239	0.125	1.916	0.055	0.239	0.143
adhd8 t4	2.488	0.175	14.241	0.000	2.488	1.485
adhd9 t1	-2.119	0.174	-12.186	0.000	-2.119	-1.076
adhd9 t2	-0.756	0.144	-5.240	0.000	-0.756	-0.384
adhd9 t3	0.629	0.149	4.223	0.000	0.629	0.320
adhd9 t4	2.488	0.167	14.907	0.000	2.488	1.264
adhd10 t1	-2.162	0.164	-13.196	0.000	-2.162	-1.284
adhd10 t2	-0.662	0.125	-5.313	0.000	-0.662	-0.393
adhd10 t3	0.478	0.127	3.757	0.000	0.478	0.284
adhd10 t4	2.162	0.160	13.512	0.000	2.162	1.284
adhd11 t1	-2.213	0.163	-13.539	0.000	-2.213	-1.284
adhd11 t2	-0.859	0.128	-6.709	0.000	-0.859	-0.498
adhd11 t3	0.412	0.129	3.187	0.001	0.412	0.239
adhd11 t4	2.395	0.179	13.345	0.000	2.395	1.389
adhd12 t1	-1.755	0.147	-11.963	0.000	-1.755	-0.961
adhd12 t2	-0.601	0.135	-4.443	0.000	-0.601	-0.329
adhd12 t3	0.469	0.137	3.429	0.001	0.469	0.257

adhd12 t4	2.345	0.175	13.419	0.000	2.345	1.284
adhd13 t1	-2.072	0.168	-12.346	0.000	-2.072	-1.124
adhd13 t2	-0.607	0.135	-4.508	0.000	-0.607	-0.329
adhd13 t3	0.376	0.138	2.722	0.006	0.376	0.204
adhd13 t4	2.297	0.161	14.283	0.000	2.297	1.245
adhd14 t1	-2.014	0.161	-12.486	0.000	-2.014	-1.124
adhd14 t2	-0.606	0.132	-4.587	0.000	-0.606	-0.338
adhd14 t3	0.461	0.135	3.416	0.001	0.461	0.257
adhd14 t4	2.490	0.174	14.298	0.000	2.490	1.389
adhd15 t1	-2.050	0.153	-13.403	0.000	-2.050	-1.345
adhd15 t2	-0.613	0.115	-5.330	0.000	-0.613	-0.403
adhd15 t3	0.405	0.113	3.582	0.000	0.405	0.266
adhd15 t4	1.617	0.137	11.799	0.000	1.617	1.061
adhd16 t1	-2.022	0.152	-13.298	0.000	-2.022	-1.209
adhd16 t2	-0.385	0.124	-3.108	0.002	-0.385	-0.230
adhd16 t3	0.505	0.126	4.011	0.000	0.505	0.302
adhd16 t4	2.215	0.165	13.457	0.000	2.215	1.324
adhd17 t1	-1.847	0.149	-12.378	0.000	-1.847	-1.124
adhd17 t2	-0.078	0.121	-0.645	0.519	-0.078	-0.047
adhd17 t3	0.586	0.124	4.707	0.000	0.586	0.356
adhd17 t4	2.359	0.164	14.357	0.000	2.359	1.435
adhd18 t1	-1.739	0.136	-12.741	0.000	-1.739	-1.245
adhd18 t2	-0.409	0.104	-3.931	0.000	-0.409	-0.293
adhd18 t3	0.199	0.103	1.931	0.054	0.199	0.143
adhd18 t4	1.820	0.132	13.817	0.000	1.820	1.303
inter_mjrtty t1	-0.517	0.080	-6.434	0.000	-0.517	-0.498

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.adhd1	1.000				1.000	0.408
.adhd2	1.000				1.000	0.455
.adhd3	1.000				1.000	0.318
.adhd4	1.000				1.000	0.370
.adhd5	1.000				1.000	0.338
.adhd6	1.000				1.000	0.423
.adhd7	1.000				1.000	0.311
.adhd8	1.000				1.000	0.356
.adhd9	1.000				1.000	0.258
.adhd10	1.000				1.000	0.352
.adhd11	1.000				1.000	0.336
.adhd12	1.000				1.000	0.300
.adhd13	1.000				1.000	0.294
.adhd14	1.000				1.000	0.311
.adhd15	1.000				1.000	0.431
.adhd16	1.000				1.000	0.357
.adhd17	1.000				1.000	0.370
.adhd18	1.000				1.000	0.513
.inter_majority	1.000				1.000	0.927
ADHD	1.000				1.000	1.000

Scales y*:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
adhd1	0.639				0.639	1.000
adhd2	0.674				0.674	1.000

adhd3	0.564	0.564	1.000
adhd4	0.608	0.608	1.000
adhd5	0.581	0.581	1.000
adhd6	0.650	0.650	1.000
adhd7	0.557	0.557	1.000
adhd8	0.597	0.597	1.000
adhd9	0.508	0.508	1.000
adhd10	0.594	0.594	1.000
adhd11	0.580	0.580	1.000
adhd12	0.547	0.547	1.000
adhd13	0.542	0.542	1.000
adhd14	0.558	0.558	1.000
adhd15	0.656	0.656	1.000
adhd16	0.598	0.598	1.000
adhd17	0.608	0.608	1.000
adhd18	0.716	0.716	1.000
inter_majority	0.963	0.963	1.000

```
parameterEstimates(
  fit_sem_inter,
  standardized = TRUE,
  ci = TRUE
) %>%
  filter(op == "~")
```

```
      lhs op  rhs  est   se    z pvalue ci.lower ci.upper std.lv
1 inter_majority ~ ADHD 0.282 0.082 3.441 0.001    0.121    0.442 0.282
  std.all
1    0.271
```

Structural model: latent ADHD predicting intrasexual-competition binary outcome

```
dat_sem_intra <- dat %>%
  select(all_of(adhd_items), intra_majority) %>%
  drop_na() %>%
  mutate(across(everything(), ordered))

sem_intra_model <- paste0(
  "ADHD =~ ",
  paste(adhd_items, collapse = " + "),
  "\nintra_majority ~ ADHD"
)

fit_sem_intra <- sem(
  sem_intra_model,
  data = dat_sem_intra,
  ordered = c(adhd_items, "intra_majority"),
  estimator = "WLSMV",
  parameterization = "theta",
  std.lv = TRUE
)

summary(
```

```
fit_sem_intra,
fit.measures = TRUE,
standardized = TRUE
)
```

lavaan 0.6-21 ended normally after 32 iterations

Estimator	DWLS
Optimization method	NLMINB
Number of model parameters	92

Number of observations	291
------------------------	-----

Model Test User Model:

	Standard	Scaled
Test Statistic	309.880	480.633
Degrees of freedom	152	152
P-value (Unknown)	NA	0.000
Scaling correction factor		0.738
Shift parameter		60.747
simple second-order correction		

Model Test Baseline Model:

Test statistic	53857.185	15701.999
Degrees of freedom	171	171
P-value	NA	0.000
Scaling correction factor		3.457

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.997	0.979
Tucker-Lewis Index (TLI)	0.997	0.976
Robust Comparative Fit Index (CFI)		0.902
Robust Tucker-Lewis Index (TLI)		0.890

Root Mean Square Error of Approximation:

RMSEA	0.060	0.086
90 Percent confidence interval - lower	0.050	0.078
90 Percent confidence interval - upper	0.069	0.095
P-value H ₀ : RMSEA ≤ 0.050	0.046	0.000
P-value H ₀ : RMSEA ≥ 0.080	0.000	0.888
Robust RMSEA		0.104
90 Percent confidence interval - lower		0.094
90 Percent confidence interval - upper		0.114
P-value H ₀ : Robust RMSEA ≤ 0.050		0.000
P-value H ₀ : Robust RMSEA ≥ 0.080		1.000

Standardized Root Mean Square Residual:

SRMR	0.053	0.053
------	-------	-------

Parameter Estimates:

Parameterization	Theta
Standard errors	Robust.sem
Information	Expected
Information saturated (h1) model	Unstructured

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
ADHD =~						
adhd1	1.204	0.094	12.863	0.000	1.204	0.769
adhd2	1.095	0.090	12.133	0.000	1.095	0.739
adhd3	1.463	0.108	13.499	0.000	1.463	0.826
adhd4	1.304	0.096	13.642	0.000	1.304	0.794
adhd5	1.399	0.102	13.673	0.000	1.399	0.814
adhd6	1.168	0.089	13.102	0.000	1.168	0.760
adhd7	1.489	0.109	13.605	0.000	1.489	0.830
adhd8	1.344	0.098	13.730	0.000	1.344	0.802
adhd9	1.696	0.122	13.868	0.000	1.696	0.861
adhd10	1.356	0.097	14.019	0.000	1.356	0.805
adhd11	1.405	0.100	14.052	0.000	1.405	0.815
adhd12	1.529	0.113	13.570	0.000	1.529	0.837
adhd13	1.550	0.116	13.357	0.000	1.550	0.840
adhd14	1.488	0.108	13.812	0.000	1.488	0.830
adhd15	1.150	0.091	12.631	0.000	1.150	0.755
adhd16	1.341	0.103	12.998	0.000	1.341	0.802
adhd17	1.305	0.098	13.281	0.000	1.305	0.794
adhd18	0.975	0.083	11.748	0.000	0.975	0.698

Regressions:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
intra_majority ~						
ADHD	-0.282	0.082	-3.441	0.001	-0.282	-0.271

Thresholds:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
adhd1 t1	-1.810	0.146	-12.363	0.000	-1.810	-1.157
adhd1 t2	-0.872	0.120	-7.278	0.000	-0.872	-0.557
adhd1 t3	0.210	0.116	1.805	0.071	0.210	0.134
adhd1 t4	2.324	0.154	15.130	0.000	2.324	1.485
adhd2 t1	-1.793	0.142	-12.668	0.000	-1.793	-1.209
adhd2 t2	-0.542	0.110	-4.938	0.000	-0.542	-0.365
adhd2 t3	0.342	0.111	3.089	0.002	0.342	0.230
adhd2 t4	1.619	0.135	11.972	0.000	1.619	1.092
adhd3 t1	-1.828	0.156	-11.703	0.000	-1.828	-1.031
adhd3 t2	-0.408	0.129	-3.168	0.002	-0.408	-0.230
adhd3 t3	0.713	0.138	5.153	0.000	0.713	0.403
adhd3 t4	2.310	0.171	13.473	0.000	2.310	1.303
adhd4 t1	-1.901	0.149	-12.724	0.000	-1.901	-1.157
adhd4 t2	-0.677	0.120	-5.633	0.000	-0.677	-0.412
adhd4 t3	0.292	0.122	2.382	0.017	0.292	0.178
adhd4 t4	1.957	0.151	12.997	0.000	1.957	1.191
adhd5 t1	-2.048	0.152	-13.454	0.000	-2.048	-1.191

adhd5 t2	-0.856	0.130	-6.604	0.000	-0.856	-0.498
adhd5 t3	0.472	0.128	3.688	0.000	0.472	0.275
adhd5 t4	1.773	0.144	12.279	0.000	1.773	1.031
adhd6 t1	-2.068	0.155	-13.325	0.000	-2.068	-1.345
adhd6 t2	-0.677	0.117	-5.791	0.000	-0.677	-0.440
adhd6 t3	0.273	0.114	2.401	0.016	0.273	0.178
adhd6 t4	1.779	0.133	13.416	0.000	1.779	1.157
adhd7 t1	-2.375	0.177	-13.388	0.000	-2.375	-1.324
adhd7 t2	-0.824	0.134	-6.140	0.000	-0.824	-0.459
adhd7 t3	0.429	0.134	3.197	0.001	0.429	0.239
adhd7 t4	2.075	0.171	12.121	0.000	2.075	1.157
adhd8 t1	-1.966	0.153	-12.867	0.000	-1.966	-1.174
adhd8 t2	-0.785	0.125	-6.289	0.000	-0.785	-0.469
adhd8 t3	0.239	0.125	1.916	0.055	0.239	0.143
adhd8 t4	2.488	0.175	14.241	0.000	2.488	1.485
adhd9 t1	-2.119	0.174	-12.186	0.000	-2.119	-1.076
adhd9 t2	-0.756	0.144	-5.240	0.000	-0.756	-0.384
adhd9 t3	0.629	0.149	4.223	0.000	0.629	0.320
adhd9 t4	2.488	0.167	14.907	0.000	2.488	1.264
adhd10 t1	-2.162	0.164	-13.196	0.000	-2.162	-1.284
adhd10 t2	-0.662	0.125	-5.313	0.000	-0.662	-0.393
adhd10 t3	0.478	0.127	3.757	0.000	0.478	0.284
adhd10 t4	2.162	0.160	13.512	0.000	2.162	1.284
adhd11 t1	-2.213	0.163	-13.539	0.000	-2.213	-1.284
adhd11 t2	-0.859	0.128	-6.709	0.000	-0.859	-0.498
adhd11 t3	0.412	0.129	3.187	0.001	0.412	0.239
adhd11 t4	2.395	0.179	13.345	0.000	2.395	1.389
adhd12 t1	-1.755	0.147	-11.963	0.000	-1.755	-0.961
adhd12 t2	-0.601	0.135	-4.443	0.000	-0.601	-0.329
adhd12 t3	0.469	0.137	3.429	0.001	0.469	0.257
adhd12 t4	2.345	0.175	13.419	0.000	2.345	1.284
adhd13 t1	-2.072	0.168	-12.346	0.000	-2.072	-1.124
adhd13 t2	-0.607	0.135	-4.508	0.000	-0.607	-0.329
adhd13 t3	0.376	0.138	2.722	0.006	0.376	0.204
adhd13 t4	2.297	0.161	14.283	0.000	2.297	1.245
adhd14 t1	-2.014	0.161	-12.486	0.000	-2.014	-1.124
adhd14 t2	-0.606	0.132	-4.587	0.000	-0.606	-0.338
adhd14 t3	0.461	0.135	3.416	0.001	0.461	0.257
adhd14 t4	2.490	0.174	14.298	0.000	2.490	1.389
adhd15 t1	-2.050	0.153	-13.403	0.000	-2.050	-1.345
adhd15 t2	-0.613	0.115	-5.330	0.000	-0.613	-0.403
adhd15 t3	0.405	0.113	3.582	0.000	0.405	0.266
adhd15 t4	1.617	0.137	11.799	0.000	1.617	1.061
adhd16 t1	-2.022	0.152	-13.298	0.000	-2.022	-1.209
adhd16 t2	-0.385	0.124	-3.108	0.002	-0.385	-0.230
adhd16 t3	0.505	0.126	4.011	0.000	0.505	0.302
adhd16 t4	2.215	0.165	13.457	0.000	2.215	1.324
adhd17 t1	-1.847	0.149	-12.378	0.000	-1.847	-1.124
adhd17 t2	-0.078	0.121	-0.645	0.519	-0.078	-0.047
adhd17 t3	0.586	0.124	4.707	0.000	0.586	0.356
adhd17 t4	2.359	0.164	14.357	0.000	2.359	1.435
adhd18 t1	-1.739	0.136	-12.741	0.000	-1.739	-1.245
adhd18 t2	-0.409	0.104	-3.931	0.000	-0.409	-0.293
adhd18 t3	0.199	0.103	1.931	0.054	0.199	0.143

adhd18 t4	1.820	0.132	13.817	0.000	1.820	1.303
intra_mjrty t1	0.517	0.080	6.434	0.000	0.517	0.498

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.adhd1	1.000				1.000	0.408
.adhd2	1.000				1.000	0.455
.adhd3	1.000				1.000	0.318
.adhd4	1.000				1.000	0.370
.adhd5	1.000				1.000	0.338
.adhd6	1.000				1.000	0.423
.adhd7	1.000				1.000	0.311
.adhd8	1.000				1.000	0.356
.adhd9	1.000				1.000	0.258
.adhd10	1.000				1.000	0.352
.adhd11	1.000				1.000	0.336
.adhd12	1.000				1.000	0.300
.adhd13	1.000				1.000	0.294
.adhd14	1.000				1.000	0.311
.adhd15	1.000				1.000	0.431
.adhd16	1.000				1.000	0.357
.adhd17	1.000				1.000	0.370
.adhd18	1.000				1.000	0.513
.intra_majority	1.000				1.000	0.927
ADHD	1.000				1.000	1.000

Scales y*:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
adhd1	0.639				0.639	1.000
adhd2	0.674				0.674	1.000
adhd3	0.564				0.564	1.000
adhd4	0.608				0.608	1.000
adhd5	0.581				0.581	1.000
adhd6	0.650				0.650	1.000
adhd7	0.557				0.557	1.000
adhd8	0.597				0.597	1.000
adhd9	0.508				0.508	1.000
adhd10	0.594				0.594	1.000
adhd11	0.580				0.580	1.000
adhd12	0.547				0.547	1.000
adhd13	0.542				0.542	1.000
adhd14	0.558				0.558	1.000
adhd15	0.656				0.656	1.000
adhd16	0.598				0.598	1.000
adhd17	0.608				0.608	1.000
adhd18	0.716				0.716	1.000
intra_majority	0.963				0.963	1.000

```
parameterEstimates(
  fit_sem_intra,
  standardized = TRUE,
  ci = TRUE
```

```
) %>%
  filter(op == "~")
```

```
      lhs op  rhs    est    se      z pvalue ci.lower ci.upper std.lv
1 intra_majority ~ ADHD -0.282 0.082 -3.441 0.001 -0.442 -0.121 -0.282
  std.all
1 -0.271
```

```
# Function: row mean with minimum number of valid items
row_mean_min <- function(data, cols, min_items = 1) {
  x <- data[, cols, drop = FALSE]

  # make sure columns are numeric
  x <- as.data.frame(lapply(x, function(v) as.numeric(as.character(v))))

  n_valid <- rowSums(!is.na(x))
  out <- rowMeans(x, na.rm = TRUE)

  out[n_valid < min_items] <- NA_real_
  out
}

# Function: row mean with minimum number of valid items
row_mean_min <- function(data, cols, min_items = 1) {
  x <- data[, cols, drop = FALSE]

  # make sure columns are numeric
  x <- as.data.frame(lapply(x, function(v) as.numeric(as.character(v))))

  n_valid <- rowSums(!is.na(x))
  out <- rowMeans(x, na.rm = TRUE)

  out[n_valid < min_items] <- NA_real_
  out
}
```

```
library(dplyr)

# Final analysis dataset: complete-case sample for paper
# This forces the logistic regression, CFA, reliability, and group comparison
# to use the same ADHD item-level complete-case sample.

dat_paper <- dat %>%
  filter(
    if_all(all_of(adhd_items), ~ !is.na(.x)),
    !is.na(inter_majority)
  ) %>%
  mutate(
    adhd_total = rowMeans(across(all_of(adhd_items)), na.rm = FALSE),
    adhd_inattention = rowMeans(across(all_of(inatt_items)), na.rm = FALSE),
    adhd_hyperimpulsive = rowMeans(across(all_of(hyper_items)), na.rm = FALSE)
  )

# Check final sample
```

```
n_raw <- nrow(dat)
n_final <- nrow(dat_paper)
n_removed <- n_raw - n_final

n_final
```

[1] 291

```
age_midpoint <- function(x) {
  x <- as.character(x)
  nums <- stringr::str_extract_all(x, "\\d+")

  purrr::map_dbl(nums, function(z) {
    z <- as.numeric(z)

    if (length(z) >= 2) {
      mean(z[1:2])
    } else if (length(z) == 1) {
      z[1]
    } else {
      NA_real_
    }
  })
}

dat_paper <- dat_paper %>%
  mutate(age_mid = age_midpoint(age))

age_table <- dat_paper %>%
  count(age, name = "n") %>%
  mutate(percent = round(100 * n / sum(n), 2))

knitr::kable(
  age_table,
  caption = "Age category distribution"
)
```

Age category distribution

age	n	percent
18-24	70	24.05
25-34	161	55.33
35-44	52	17.87
45-54	8	2.75

```
age_summary <- dat_paper %>%
  summarise(
    n_age = sum(!is.na(age_mid)),
    mean_age = mean(age_mid, na.rm = TRUE),
    sd_age = sd(age_mid, na.rm = TRUE),
```

```

    min_age = min(age_mid, na.rm = TRUE),
    max_age = max(age_mid, na.rm = TRUE)
  ) %>%
  mutate(across(where(is.numeric), ~ round(.x, 2)))

knitr::kable(
  age_summary,
  caption = "Approximate age summary using category midpoints"
)

```

Approximate age summary using category midpoints

n_age	mean_age	sd_age	min_age	max_age
291	29.79	6.81	21	49.5

##ADHD reliability: alpha and omega

```

# Reset item names to match your current dataset
adhd_items <- paste0("adhd", 1:18)

inatt_items <- c(
  "adhd1", "adhd2", "adhd3", "adhd4",
  "adhd7", "adhd8", "adhd9", "adhd10", "adhd11"
)

hyper_items <- c(
  "adhd5", "adhd6",
  "adhd12", "adhd13", "adhd14", "adhd15",
  "adhd16", "adhd17", "adhd18"
)

# Check that the columns exist
stopifnot(all(adhd_items %in% names(dat_paper)))
stopifnot(all(inatt_items %in% names(dat_paper)))
stopifnot(all(hyper_items %in% names(dat_paper)))

# Reliability dataset
adhd_reliability_dat <- dat_paper %>%
  select(all_of(adhd_items)) %>%
  drop_na()

# Cronbach's alpha
alpha_total <- psych::alpha(adhd_reliability_dat)
alpha_inattention <- psych::alpha(
  adhd_reliability_dat %>% select(all_of(inatt_items))
)
alpha_hyper <- psych::alpha(
  adhd_reliability_dat %>% select(all_of(hyper_items))
)

# Omega
omega_total <- psych::omega(
  adhd_reliability_dat,

```

```
  nfactors = 2,
  plot = FALSE
)

# Clean reliability table
reliability_table <- tibble(
  Scale = c(
    "ADHD total",
    "Inattention",
    "Hyperactivity / impulsivity",
    "ADHD total omega total",
    "ADHD total omega hierarchical"
  ),
  Estimate = c(
    alpha_total$total$raw_alpha,
    alpha_inattention$total$raw_alpha,
    alpha_hyper$total$raw_alpha,
    omega_total$omega.tot,
    omega_total$omega_h
  )
) %>%
  mutate(Estimate = round(Estimate, 3))

knitr::kable(
  reliability_table,
  caption = "Reliability estimates for ADHD items"
)
```

Reliability estimates for ADHD items

Scale	Estimate
ADHD total	0.964
Inattention	0.939
Hyperactivity / impulsivity	0.935
ADHD total omega total	0.967
ADHD total omega hierarchical	0.857

two factor ADHD CFA

```
dat_cfa_2factor <- dat_paper %>%
  select(all_of(adhd_items)) %>%
  drop_na() %>%
  mutate(across(everything(), ordered))

adhd_2factor_model <- '
  Inattention =~ adhd1 + adhd2 + adhd3 + adhd4 +
                adhd7 + adhd8 + adhd9 + adhd10 + adhd11

  HyperImp =~ adhd5 + adhd6 + adhd12 + adhd13 + adhd14 +
              adhd15 + adhd16 + adhd17 + adhd18
```

```
Inattention ~~ HyperImp
,

fit_adhd_2factor <- cfa(
  model = adhd_2factor_model,
  data = dat_cfa_2factor,
  ordered = adhd_items,
  estimator = "WLSMV",
  parameterization = "theta",
  std.lv = TRUE
)

fit_2factor_table <- tibble(
  Model = "Two-factor ADHD",
  chisq = fitMeasures(fit_adhd_2factor, "chisq"),
  df = fitMeasures(fit_adhd_2factor, "df"),
  CFI = fitMeasures(fit_adhd_2factor, "cfi"),
  TLI = fitMeasures(fit_adhd_2factor, "tli"),
  RMSEA = fitMeasures(fit_adhd_2factor, "rmsea"),
  SRMR = fitMeasures(fit_adhd_2factor, "srmr")
) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

knitr::kable(
  fit_2factor_table,
  caption = "Two-factor ADHD CFA model fit"
)
```

Two-factor ADHD CFA model fit

Model	chisq	df	CFI	TLI	RMSEA	SRMR
Two-factor ADHD	179.093	134	0.999	0.999	0.034	0.036

comparison

```
knitr::kable(
  fit_2factor_table,
  caption = "Two-factor ADHD CFA model fit"
)
```

Two-factor ADHD CFA model fit

Model	chisq	df	CFI	TLI	RMSEA	SRMR
Two-factor ADHD	179.093	134	0.999	0.999	0.034	0.036

```
adhd_2factor_model <- '
  Inattention =~ adhd1 + adhd2 + adhd3 + adhd4 +
                adhd7 + adhd8 + adhd9 + adhd10 + adhd11
```

```
HyperImp =~ adhd5 + adhd6 + adhd12 + adhd13 + adhd14 +
           adhd15 + adhd16 + adhd17 + adhd18

Inattention =~ HyperImp
,
knitr::kable(
  fit_2factor_table,
  caption = "Two-factor ADHD CFA model fit"
)
```

Two-factor ADHD CFA model fit

Model	chisq	df	CFI	TLI	RMSEA	SRMR
Two-factor ADHD	179.093	134	0.999	0.999	0.034	0.036

```
## Figure 1. Two-factor CFA path diagram

#| label: fig-cfa-path
#| fig.cap: "Two-factor CFA measurement model for ADHD symptoms. Inattention and Hyper
#| fig.width: 9
#| fig.height: 5
#| fig.align: center
#| message: false
#| warning: false

library(ggplot2)
library(dplyr)
library(grid)

# Observed item positions
items <- tibble(
  factor = rep(c("Inattention", "Hyperactivity / Impulsivity"), each = 9),
  item = c(
    "adhd1", "adhd2", "adhd3", "adhd4", "adhd7", "adhd8", "adhd9", "adhd10", "adhd11",
    "adhd5", "adhd6", "adhd12", "adhd13", "adhd14", "adhd15", "adhd16", "adhd17", "adh
  ),
  x = rep(seq(-4, 4, length.out = 9), 2),
  y = c(rep(1.1, 9), rep(-1.1, 9)),
  latent_x = 0,
  latent_y = c(rep(2.0, 9), rep(0.0, 9))
)

# Latent factor labels
latent <- tibble(
  name = c("Inattention", "Hyperactivity /\nImpulsivity"),
  x = c(0, 0),
  y = c(2.2, 0)
)

ggplot() +
  # Factor loading arrows
  geom_segment(
```



```

    data = items,
    aes(x = latent_x, y = latent_y, xend = x, yend = y + 0.18),
    arrow = arrow(length = unit(0.12, "inches")),
    linewidth = 0.35
  ) +

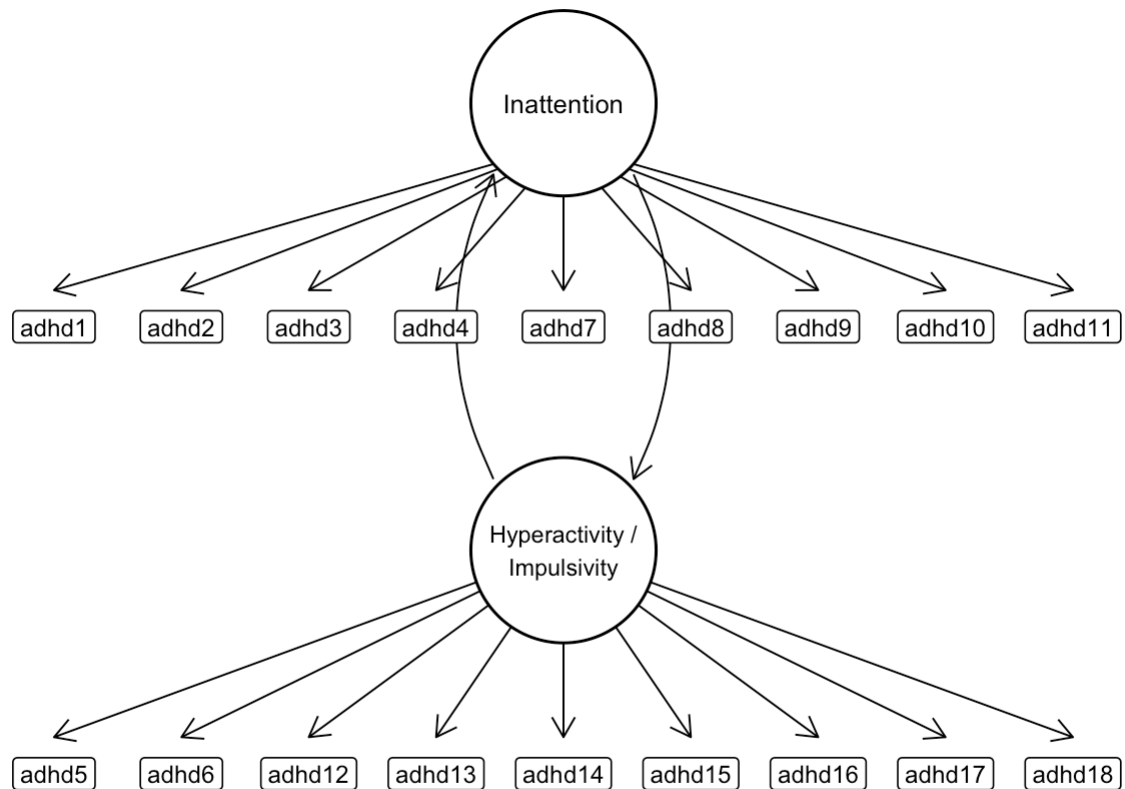
# Correlation between latent factors: two curved arrows
geom_curve(
  aes(x = 0.55, y = 1.85, xend = 0.55, yend = 0.35),
  curvature = -0.25,
  arrow = arrow(length = unit(0.12, "inches")),
  linewidth = 0.35
) +
geom_curve(
  aes(x = -0.55, y = 0.35, xend = -0.55, yend = 1.85),
  curvature = -0.25,
  arrow = arrow(length = unit(0.12, "inches")),
  linewidth = 0.35
) +

# Observed item boxes
geom_label(
  data = items,
  aes(x = x, y = y, label = item),
  fill = "white",
  label.size = 0.30,
  size = 3.2
) +

# Latent variable circles
annotate("point", x = 0, y = 2.2, shape = 21, size = 32, fill = "white", stroke = 0.8) +
annotate("text", x = 0, y = 2.2, label = "Inattention", size = 3.4) +
annotate("point", x = 0, y = 0, shape = 21, size = 32, fill = "white", stroke = 0.8) +
annotate("text", x = 0, y = 0, label = "Hyperactivity /\nImpulsivity", size = 3.1) +

coord_cartesian(xlim = c(-4.8, 4.8), ylim = c(-1.6, 2.7), clip = "off") +
theme_void()

```



```
## logistic regression
```

```
::: {.cell}
```

```
```{.r .cell-code}
```

```
dat_logit <- dat_paper %>%
 mutate(
 adhd_z = as.numeric(scale(adhd_total))
) %>%
 select(adhd_z, inter_majority) %>%
 drop_na()
```

```
m_inter_total <- glm(
 inter_majority ~ adhd_z,
 data = dat_logit,
 family = binomial(link = "logit")
)
```

```
logit_table <- broom::tidy(
 m_inter_total,
 conf.int = TRUE,
 exponentiate = FALSE
) %>%
 mutate(
 OR = exp(estimate),
 OR_low = exp(conf.low),
 OR_high = exp(conf.high)
) %>%
```

```
select(
 Predictor = term,
 B = estimate,
 SE = std.error,
 z = statistic,
 p = p.value,
 OR,
 OR_low,
 OR_high
) %>%
mutate(across(where(is.numeric), ~ round(.x, 3)))

knitr::kable(
 logit_table,
 caption = "Logistic regression predicting intersexual / ornament-like majority
choice"
)
```

Logistic regression predicting intersexual / ornament-like majority choice

Predictor	B	SE	z	p	OR	OR_low	OR_high
(Intercept)	0.839	0.131	6.400	0.000	2.314	1.798	3.008
adhd_z	0.438	0.131	3.345	0.001	1.550	1.203	2.013

...

diagnostics table

```
m_null <- glm(
 inter_majority ~ 1,
 data = dat_logit,
 family = binomial(link = "logit")
)

lrt_table <- anova(m_null, m_inter_total, test = "Chisq")

knitr::kable(
 lrt_table,
 caption = "Likelihood-ratio test comparing null model and ADHD_total model"
)
```

Likelihood-ratio test comparing null model and ADHD\_total model

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
290	359.9798	NA	NA	NA
289	348.3847	1	11.59515	0.0006612

```
diagnostic_table <- tibble(
 Diagnostic = c(
 "Model convergence",
 "Outcome type",
 "Estimator",
 "Multicollinearity"
),
 Result = c(
 ifelse(m_inter_total$converged, "Converged", "Did not converge"),
 "Binary outcome: inter_majority coded 0/1",
 "Maximum likelihood estimation via glm() with binomial logit link",
 "Not a concern because the focal model has one predictor"
)
)

knitr::kable(
 diagnostic_table,
 caption = "Model diagnostics and analytical choices"
)
```

Model diagnostics and analytical choices

Diagnostic	Result
Model convergence	Converged
Outcome type	Binary outcome: inter_majority coded 0/1
Estimator	Maximum likelihood estimation via glm() with binomial logit link
Multicollinearity	Not a concern because the focal model has one predictor

## group

```
Make sure ggplot2 is available
library(ggplot2)

Create ADHD median-split group
dat_group <- dat_paper %>%
 mutate(
 inter_majority_num = as.numeric(as.character(inter_majority)),
 adhd_group = if_else(
 adhd_total >= median(adhd_total, na.rm = TRUE),
 "Higher ADHD group",
 "Lower ADHD group"
)
) %>%
 filter(!is.na(inter_majority_num), !is.na(adhd_group))

Group summary with 95% CI
group_summary <- dat_group %>%
 group_by(adhd_group) %>%
 summarise(
```

```

 n = n(),
 proportion = mean(inter_majority_num, na.rm = TRUE),
 se = sqrt(proportion * (1 - proportion) / n),
 lower = pmax(0, proportion - 1.96 * se),
 upper = pmin(1, proportion + 1.96 * se),
 .groups = "drop"
)

Table
knitr::kable(
 group_summary %>%
 mutate(
 proportion = round(proportion, 3),
 se = round(se, 3),
 lower = round(lower, 3),
 upper = round(upper, 3)
),
 caption = "Descriptive group comparison with 95% confidence intervals"
)

```

Descriptive group comparison with 95% confidence intervals

adhd_group	n	proportion	se	lower	upper
Higher ADHD group	147	0.762	0.035	0.693	0.831
Lower ADHD group	144	0.618	0.040	0.539	0.697

```

Plot
ggplot(group_summary, aes(x = adhd_group, y = proportion)) +
 geom_col(width = 0.60) +
 geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
 scale_y_continuous(
 limits = c(0, 1),
 labels = function(x) paste0(round(x * 100), "%")
) +
 labs(
 x = NULL,
 y = "Intersexual / ornament-like majority choice",
 title = "Intersexual choice by ADHD group"
) +
 theme_minimal()

```

